

Please ensure this tracking sheet is completed as a group, as it will be used to monitor your individual participation within the team.

Team Members	Role/ Responsibility	Email				
Gilbert Muramirabagabo	Data preprocessing, eigen decomposition, variance sorting, visualization (PC space), benchmarking, GitHub submission	g.muramirab@alustudent.com				
Kevine Uwisanga	EDA, standardization, covariance matrix, dynamic component selection, visualization (original space), optimization, documentation	k.uwisanga@alustudent.com				

Task Allocation and Tracker

	Task Allocated	Assigned Member	Deadline	Completion Status (Not Started/In Progress/Completed/Did not participate)	Reviewed by Team? (Yes/No)	Comments
	Assigned MemberData collection & preprocessing (find dataset, handle missing values, encode categorical features)	Gilbert Muramirabagabo	09/06/2026	Completed	Yes	
	Exploratory data analysis & feature standardization	Kevine Uwisanga	09/06/2026	Completed	Yes	
	PCA implementation — covariance matrix & eigen decomposition (NumPy only)	Gilbert Muramirabagabo	09/06/2026	Completed	Yes	
	Explained variance calculation, eigenvalue sorting & dynamic component selection	Kevine Uwisanga	09/06/2026	Completed	Yes	
	Visualizations, original feature space & PC space plots (before and after PCA)	Gilbert Muramirabagabo	09/06/2026	Completed	Yes	
	Written analysis, component selection justification, tradeoffs & information loss	Kevine Uwisanga	09/06/2026	Completed	Yes	
	Performance optimization, benchmarking & final submission (notebook output check, PDF, GitHub)	Gilbert Muramirabagabo	09/06/2026	Completed	Yes	

Meeting Attendance Log

Meeting Date	Agenda	Facilitator	Attendees	Key Discussion Points	Next Steps
08/06/2026	Agree on dataset and divide the tasks	Gilbert Muramirabagabo	All members	Reviewed the assignment rubric, agreed on a dataset, and divided the 7 tasks between us. Set internal deadlines to guide the work ahead.	Downloaded, shared the dataset and confirmed we understood our assigned tasks before starting any work.
09/06/2026	Implementation check-in	Kevine Uwisanga	All members	Reviewed preprocessing, PCA implementation, and both visualizations to ensure correctness. Flagged and resolved any blockers before the final push.	We fixed any issues flagged during the review and shared updated work with the other for a quick check.
10/06/2026	Final review & submission	Gilbert Muramirabagabo	All members	Reviewed the written justification, ran all notebook cells, and verified every output is visible. Filled the contribution sheet, combined PDFs, and submitted the PDF file with all requirements.	We confirmed the GitHub link opens correctly and the submitted PDF contains all required pages.